

CEng 111 – Fall 2026
Week 1b

Credit: Some slides are from the “Invitation to Computer Science” book by G. M. Schneider, J. L. Gersting and some from the “Digital Design” book by M. M. Mano and M. D. Ciletti.



METU Computer Engineering

Content

Previously on CENG111!

Catalog Description:

Introduction to the fundamentals of computer systems, including computer organization, operating systems, language processors and user interfaces.Introduction to algorithms and programming. Reasoning informally about the correctness and efficiency of programs. A functional programming language will be used for practical work.

- First part of the course:
  - What is CS, computing, computation, algorithm?
  - Inside of a Computer: CPU, Memory, Gates, Boolean Logic.
  - Von Neumann architecture
  - Etc.
- Second part of the course:
  - Programming concepts with Python.

Previously on CENG111!

Grading

Grading:

Midterm	%27
Lab Exams (3-4)	%30
Von Neumann Machine Exam	%5
Participation	%3
Final	%35

- Most assignments will be programming assignments (some could be exercises to be written up).
- Laboratory schedule will be announced later. Keep watching.
- Starting with the Weeks 4-5 the laboratory sessions will go in parallel with the lectures. You will be asked to solve problems by using the ideas introduced in the lectures of that week. Earlier lab sessions will build your computing/communicating literacy.
- Attending the lab sessions is mandatory. Missing two lab sessions will result in a grade of NA for the course.



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Previously on CENG111!

Motivation

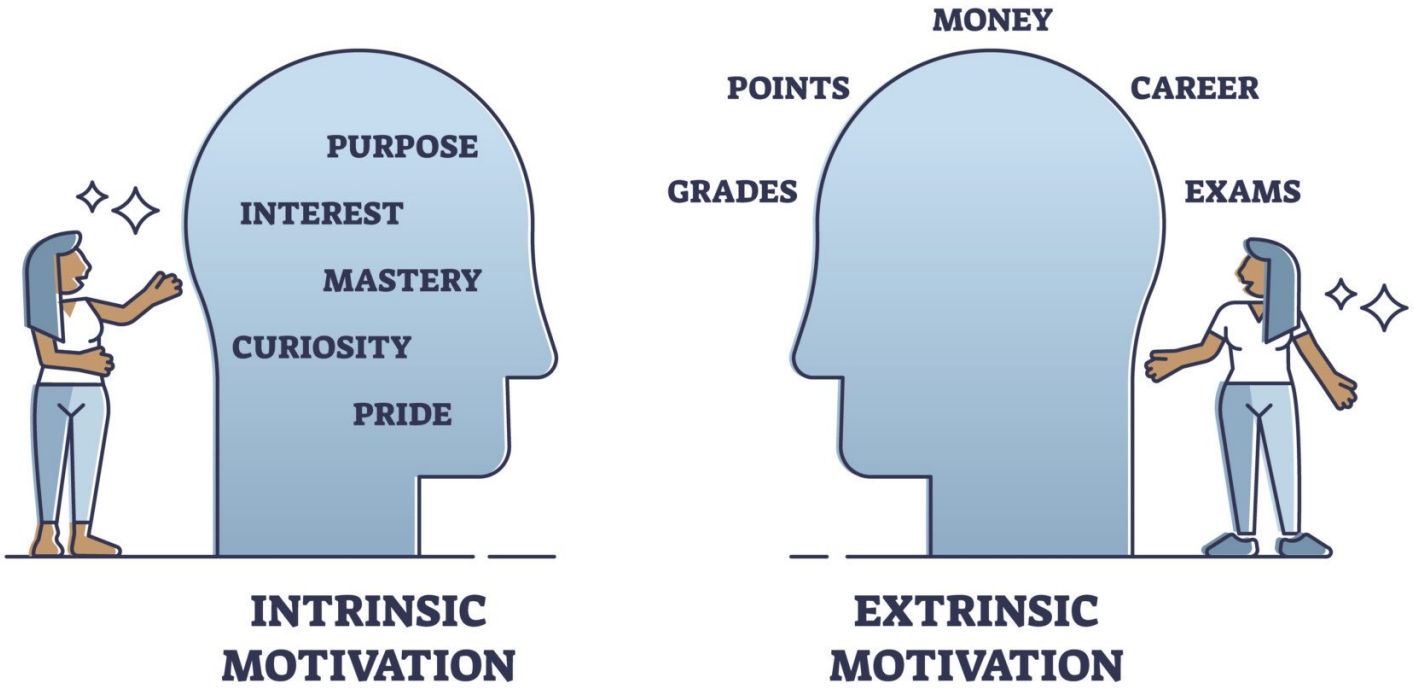


Figure: https://www.simplypsychology.org/differences-between-extrinsic-and-intrinsic-motivation.html

Computer Science is the study of algorithms (= methods)

Previously on CENG111!

including:

1. their formal and mathematical properties
2. their hardware realizations
3. their linguistic realizations
4. their applications

From “Invitation to Computer Science”

Measure the height of a tall building with a barometer

Previously on CENG111!



- What would be your answer?
- One student answered:
  - "I would tie the barometer to a rope, hang it down from the top of the building to the bottom and measure the length of the rope!"
  - Of course, the instructor rejects the answer since it doesn't include any "physics"

Check the following for two different versions of the `legend':
http://www.snopes.com/college/exam/barometer.asp

Previously on CENG111!

Analyze/Compare Algorithms
= Pros and Cons of Algorithms =

What are the disadvantages of these?

1. Drop the barometer from the top, and measure the time it takes to reach the ground.
2. Make a pendulum and time its period wrt. the top and the bottom of the building.
3. Walk down the stairs marking "barometer units" on the wall.
4. Measure its shadow and the buildings shadow. Workout the height of the building from barometer's height.

Cons:

1. You lose the barometer; it breaks.
2. You need a loooong rope and how long that rope is going to be depends on the answer to the question.
3. It takes too long. It is too tiring.
4. What if there is no sun?



Today

- Algorithm
- Computation
- Computing



Administrative Notes

- CENG account details emailed to your METU email addresses
- Fill in your schedules (deadline: 9 October)

Where does the word ‘algorithm’ come from?

- From a Persian mathematician, astronomer and geographer: Mohammed ibn-Musa al-Khwarizmi
- “Algorithmi” is the latin form of his name
- He contributed to science by
  - Decimal positional number system

    (e.g., $32 = 10^1x3 + 10^0x2$)
  - Presented the first systematic solutions to linear and quadratic equations
- In fact, the word “Algebra” comes from one of his operators (al-jabr: subtracting a number from both sides of an equation) for solving equations



Mohammed ibn-Musa al-Khwarizmi (780-850)

Source: Wikipedia

Where does the word 'algorithm' come from? (cont'd)

- al-Khwarizmi reduced equations to one of the following six forms by using al-jabr (in Arabic: restoring, completion):
  - squares equal roots ($ax^2 = bx$)
  - squares equal number ($ax^2 = c$)
  - roots equal number ($bx = c$)
  - squares and roots equal number ($ax^2 + bx = c$)
  - squares and number equal roots ($ax^2 + c = bx$)
  - roots and number equal squares ($bx + c = ax^2$)
- For example, $x^2 = 40x - 4x^2$ is reduced to $5x^2 = 40x$. From this reduced form, it is easily deducible that the variable is either 0 or 8.


كتاب الجبر والمقابلة

al-Khwarizmi's Compendious Book on Calculation by Completion and Balancing

Source: Wikipedia

What does ‘algorithm’ mean?

- “A procedure or formula for solving a problem”
- “A set of instructions to be followed to solve a problem”
- “an effective method expressed as a finite list of well-defined instructions for calculating a function”
- “step-by-step procedure for calculations”

A formal definition of algorithm

- "Starting from an initial state and initial input (perhaps empty), the instructions describe a computation that, when executed, will proceed through a finite number of well-defined successive states, eventually producing "output" and terminating at a final ending state."

What is an algorithm?

- An algorithm is a list that looks like
  - STEP 1: Do something
  - STEP 2: Do something
  - STEP 3: Do something
  - . .
  - . .
  - . .
  - STEP N: Stop, you are finished

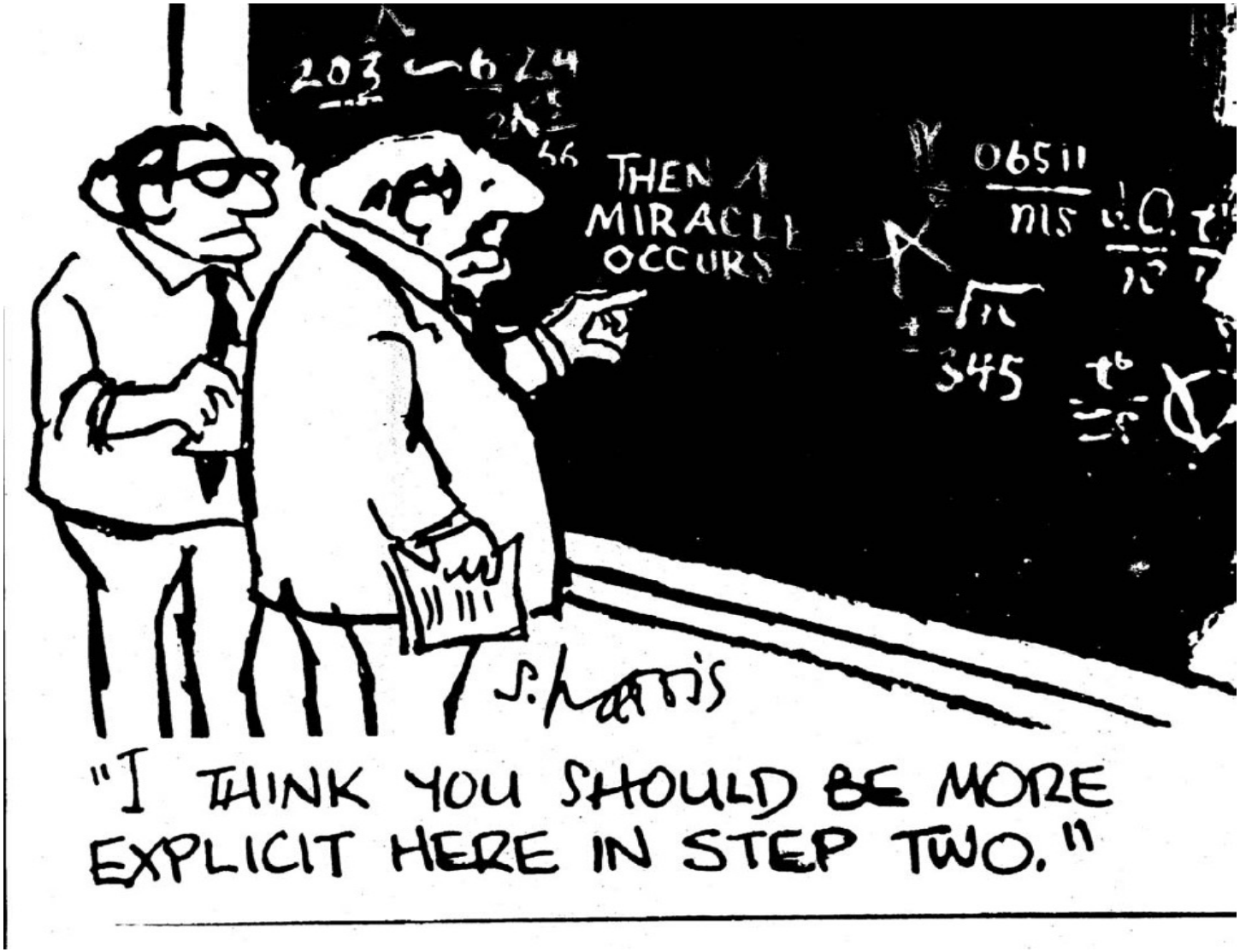
From “Invitation to Computer Science”



Valid Operations in Algorithms

- **Sequential** – simple well-defined task, usually declarative sentence.
- **Conditional-** "ask a question and select the next operation on the basis of the answer to the question – usually an "if-then" or "if then else"
- **Iterative-** "looping" instructions – repeat a set of instructions

From "Invitation to Computer Science"



From "Invitation to Computer Science"



https://www.youtube.com/watch?v=FN2RM-CHkuI



Algorithms

- We use them all the time.
- Can you give examples?
  - Following directions
  - Recording a DVD
  - Adding two numbers
  - Finding Greatest Common Divisor
  - …

From "Invitation to Computer Science"

An example algorithm from our daily lives

Algorithm for Shampooing Your Hair

STEP	OPERATION
1	Wet your hair
2	Set the value of *WashCount* to 0
3	Repeat steps 4 through 6 until the value of *WashCount* equals 2
4	Lather your hair
5	Rinse your hair
6	Add 1 to the value of *WashCount*
7	Stop, you have finished shampooing your hair

From "Invitation to Computer Science"

An example algorithm

METU Computer Engineering

Algorithm for Adding Two *m*-Digit Numbers

Given: $m \geq 1$ and two positive numbers each containing m digits, $a_{m-1}\, a_{m-2,} \ldots a_0$ and $b_{m-1}\, b_{m-2,} \ldots b_0$

Wanted: $c_m c_{m-1}\, c_{m-2} \ldots c_0$, where $c_m c_{m-1}\, c_{m-2} \ldots c_0 = (a_{m-1}\, a_{m-2} \ldots a_0) + (b_{m-1}\, b_{m-2} \ldots b_0)$

Algorithm:

Step 1 Set the value of *carry* to 0.

Step 2 Set the value of *i* to 0.

Step 3 While the value of *i* is less than or equal to $m - 1$, repeat the instructions in steps 4 through 6.

Step 4 Add the two digits a_i and b_i to the current value of *carry* to get c_i.

Step 5 If $c_i \geq 10$, then reset c_i to $(c_i - 10)$ and reset the value of *carry* to 1; otherwise, set the new value of *carry* to 0.

Step 6 Add 1 to *i*, effectively moving one column to the left.

Step 7 Set c_m to the value of *carry*.

Step 8 Print out the final answer, $c_m\, c_{m-1}\, c_{m-2} \ldots c_0$.

Step 9 Stop.

From "Invitation to Computer Science"

How to represent algorithms

- Pseudo-code

Algorithm for Adding Two *m*-Digit Numbers

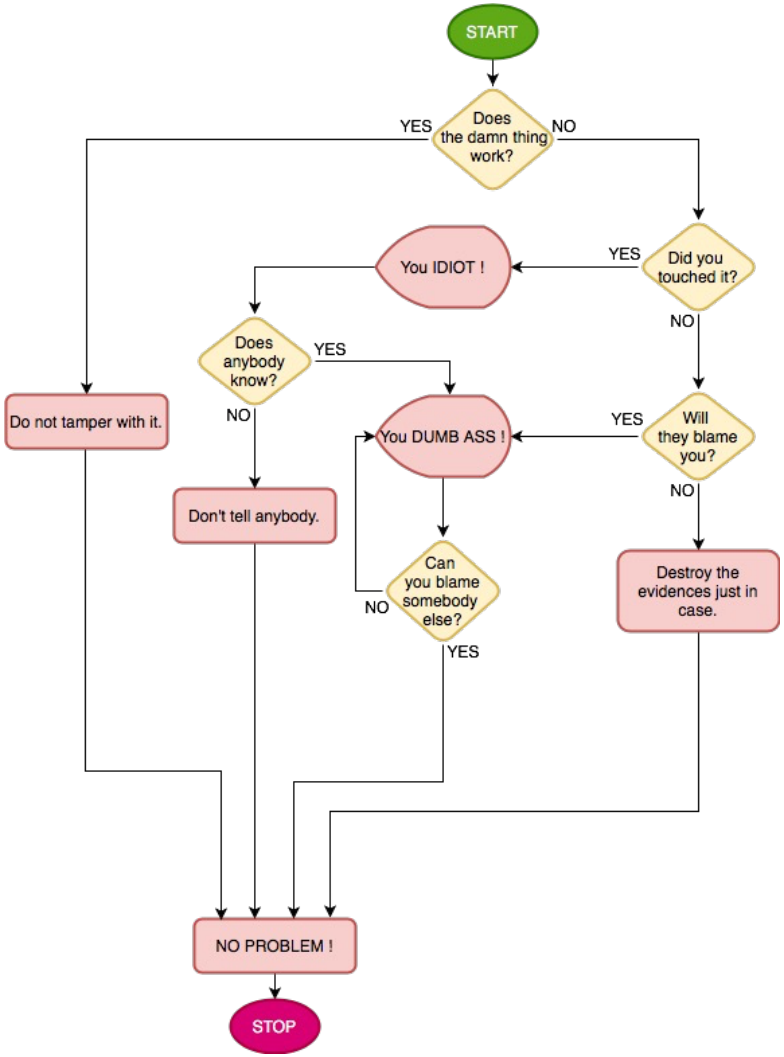
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Step 9 Stop.

How to represent algorithms

- Flow-charts





Why are algorithms important?

- If we can specify an algorithm to solve a problem then we can automate its solution.
- No algorithm => No software => No automation!

From "Invitation to Computer Science"



Can we find algorithms to all problems?

NO!

- There are problems which have no generalized solutions – unsolvable or intractable
- Some with an algorithm would take so long to execute that the algorithm is useless
- Some problems we have not yet discovered an algorithm for

From "Invitation to Computer Science"

A formal definition of algorithm

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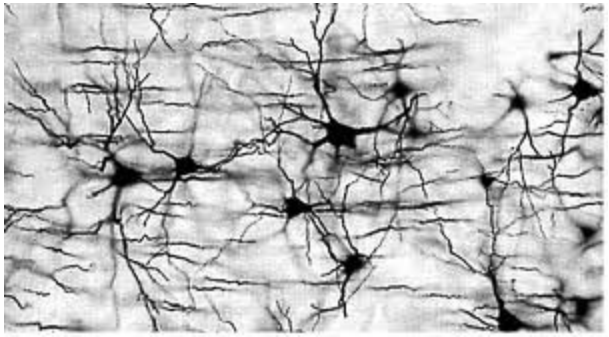
“Computation”

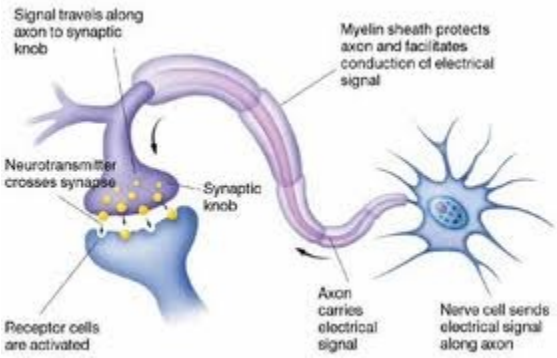
- Digital vs. analog computation
- Sequential vs. parallel computation
- Batch vs. interactive computation
- Evolutionary, molecular, quantum computation
- “Physical computation” / “Digital Physics”
  - ‘The whole universe is itself a computation’

Computation in our brain



- Highly-connected network of neurons.
- How many neurons?
  - Approx. 10^{11} neurons and 10^{14} synapses.
- How do they transmit information?
  - Using nothing else than charged molecules.





Computation in our brain (cont'd)

- Each neuron gets input and produces an output using an "activation function"

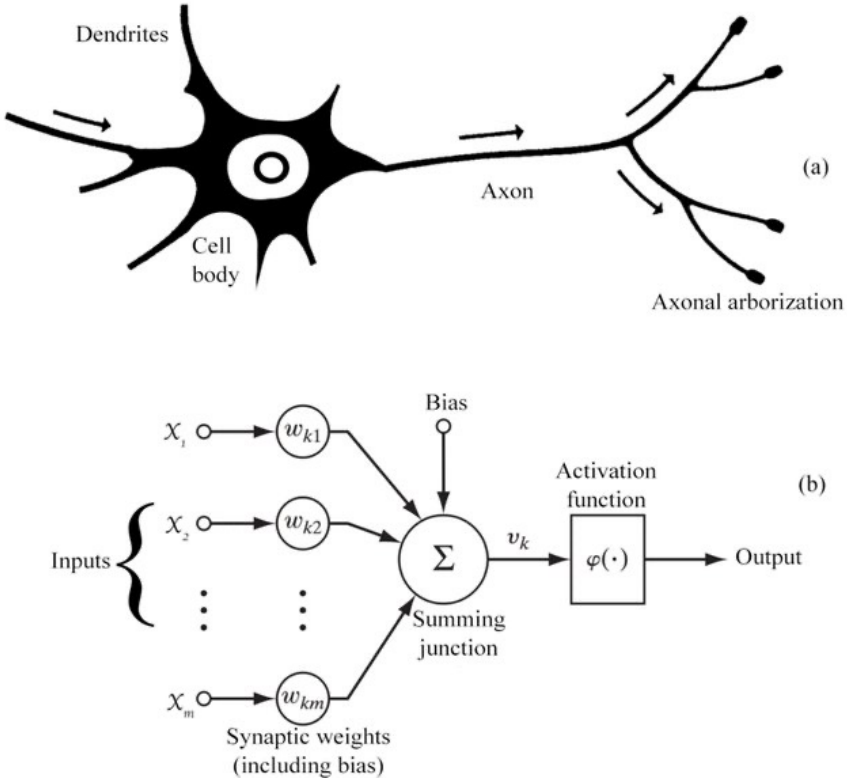
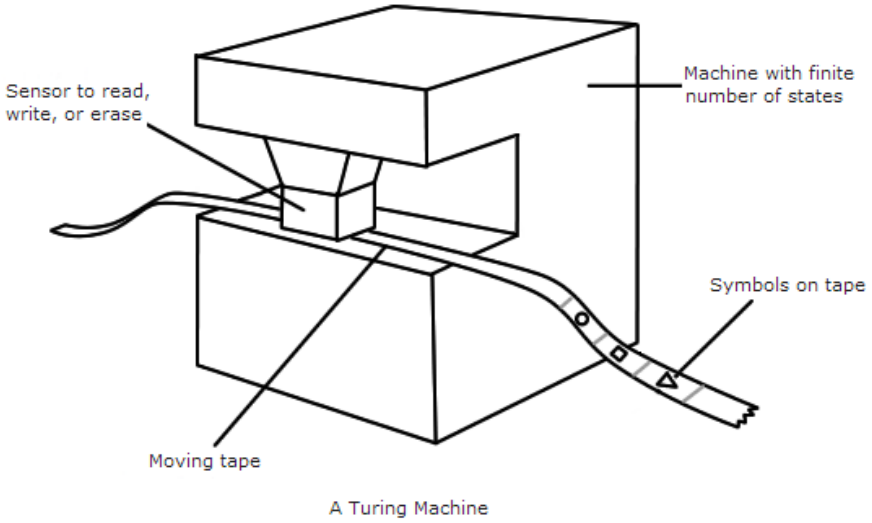
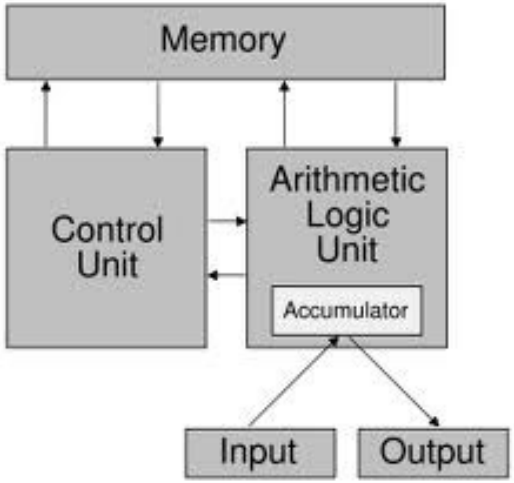


Fig: https://www.researchgate.net/publication/326417061_Modeling_Course_Achievements_of_Elementary_Education_Teacher_Candidates_with_Artificial_Neural_Networks/figures?lo=1



Turing Machine



Von Neumann Architecture

DIGITAL COMPUTATION



BUT FIRST SOME HISTORICAL OVERVIEW

The Early Period: Up to 1940



- 3,000 years ago: Mathematics, logic, and numerical computation
  - Important contributions made by the Greeks, Egyptians, Babylonians, Indians, Chinese, and Persians
  - Cuneiform
  - Stone “abacus”

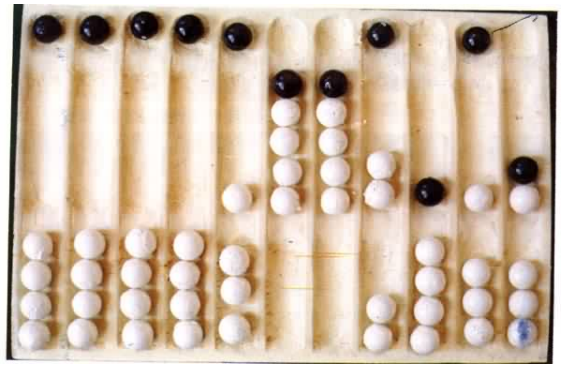
- http://www.thocp.net/slideshow/0469.htm

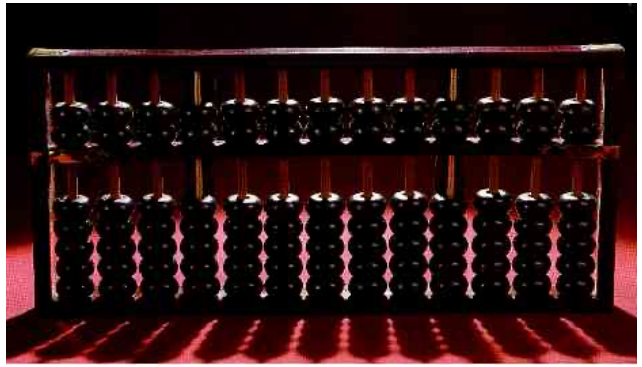
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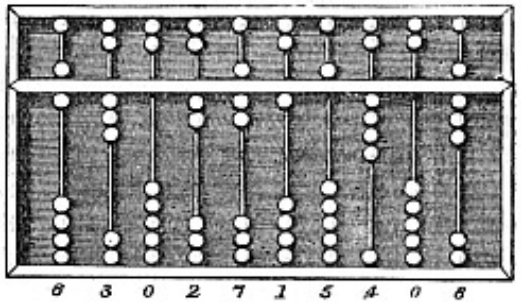
ABACUS

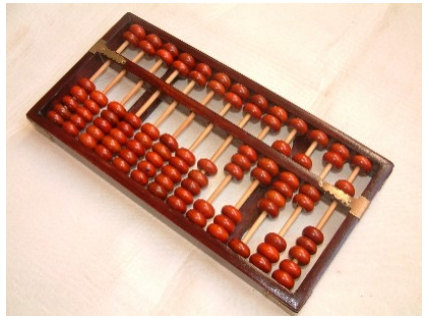
Early calculating devices

ABACUS – 2700 BC (Mesopotamia)







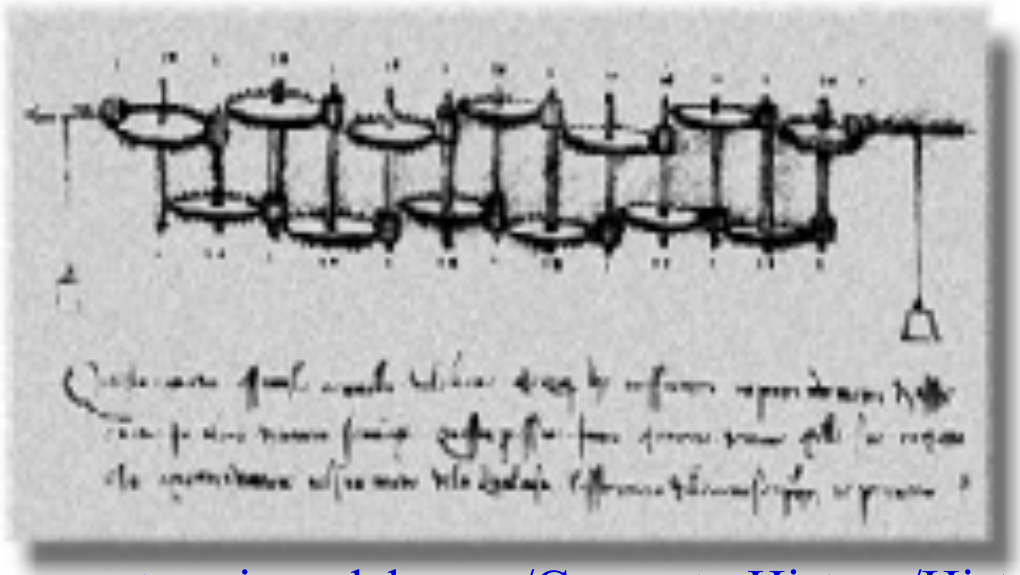


http://www.computersciencelab.com/ComputerHistory/History.htm

Slide from "Introduction to Computing"

DaVinci

- 1452-1519 Leonardo DaVinci sketched gear-driven calculating machines but none were ever built.



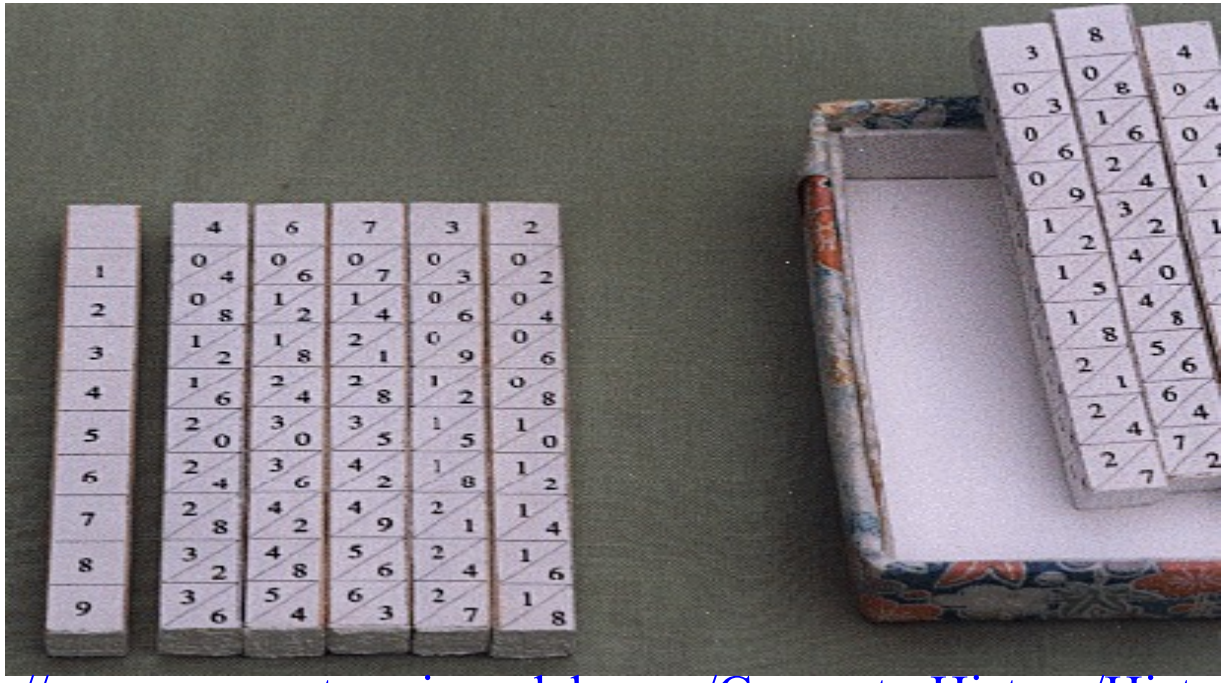
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Napier's Bones

- 1614: Logarithms
  - Invented by John Napier to simplify difficult mathematical computations

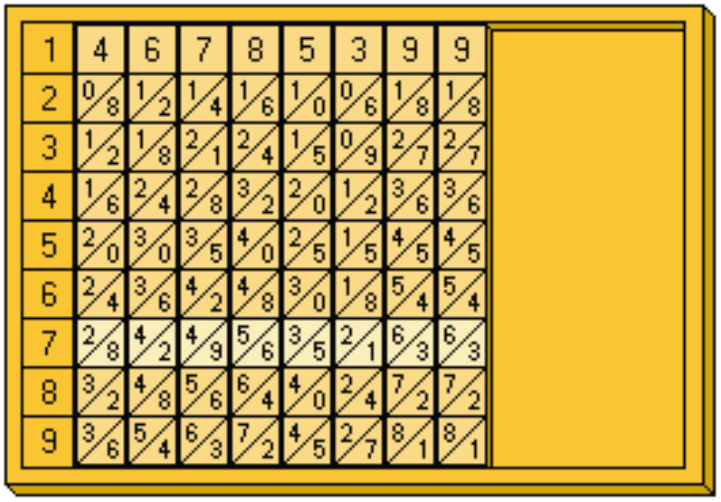
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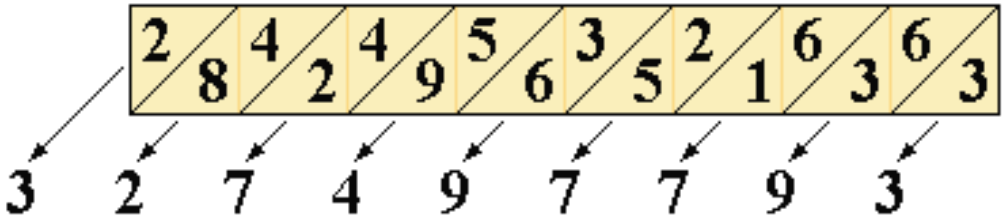


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- If you want to multiply 7 by 46785499:

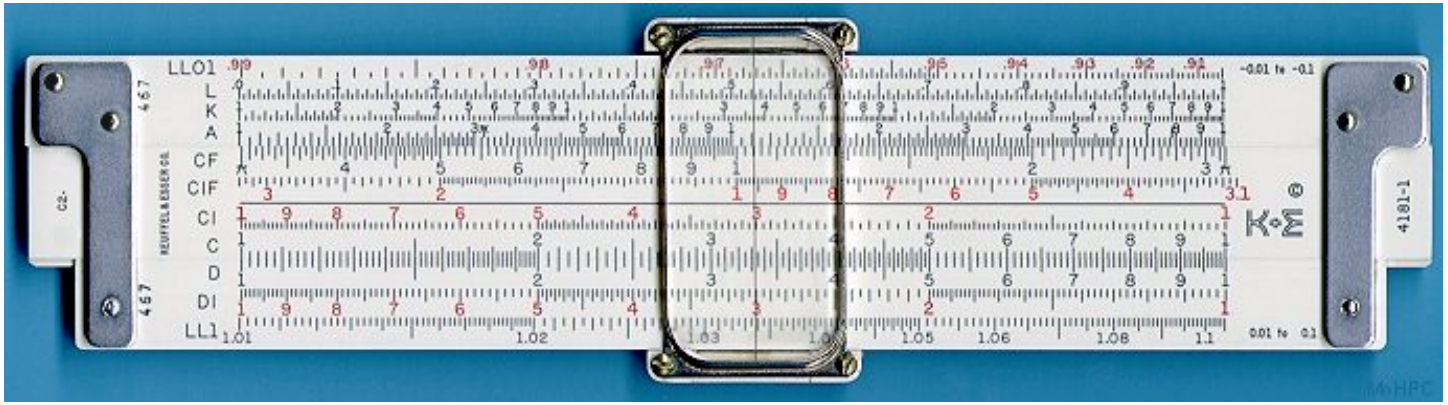




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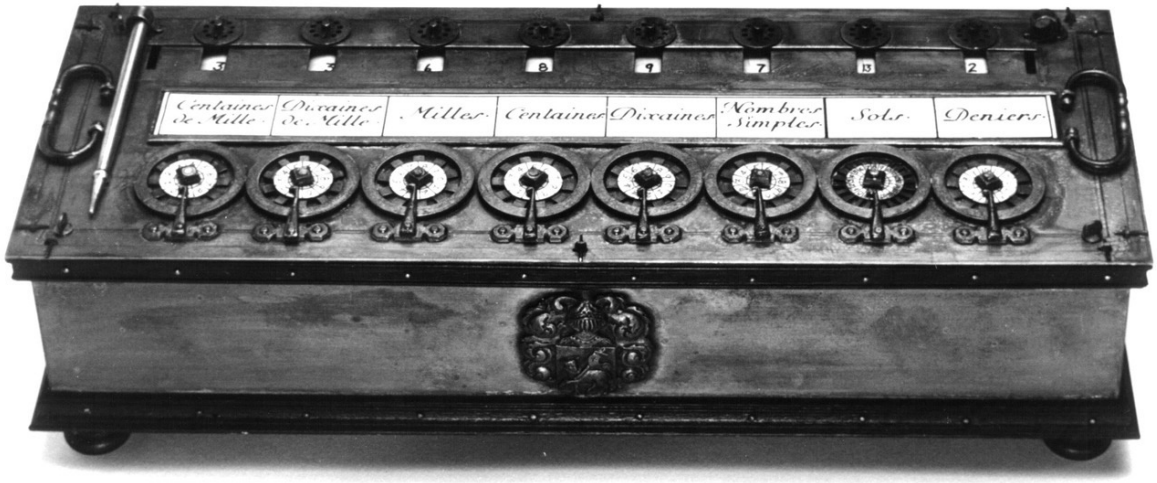
Slide Rule (slipstick)
"a mechanical analog computer"

Around 1622: First slide rule created



http://www.computersciencelab.com/ComputerHistory/History.htm

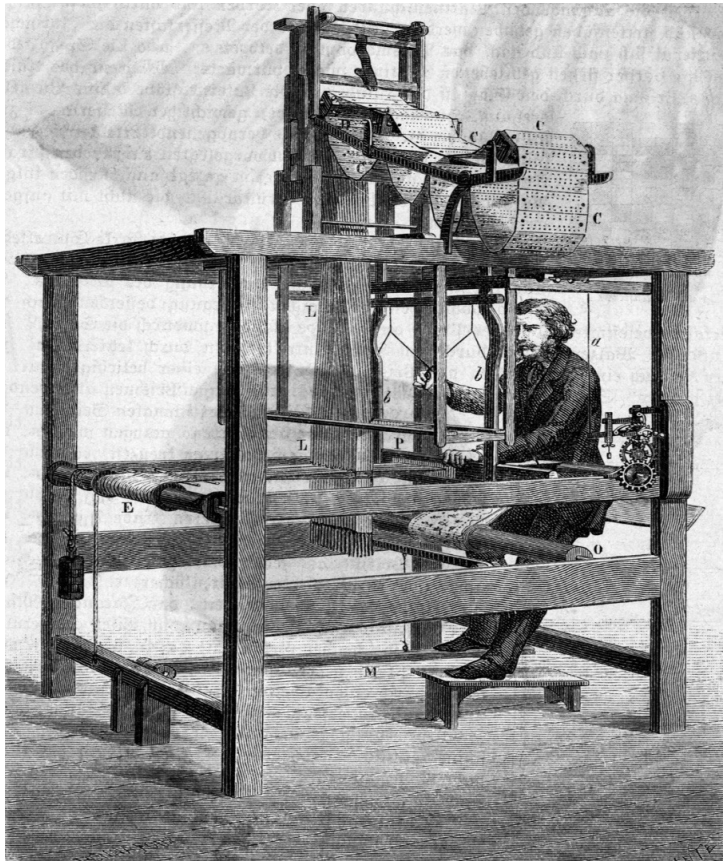
Slide from "Introduction to Computing"



The Pascaline: One of the Earliest Mechanical Calculators

Slide from “Introduction to Computing”

The Early Period: Up to 1940

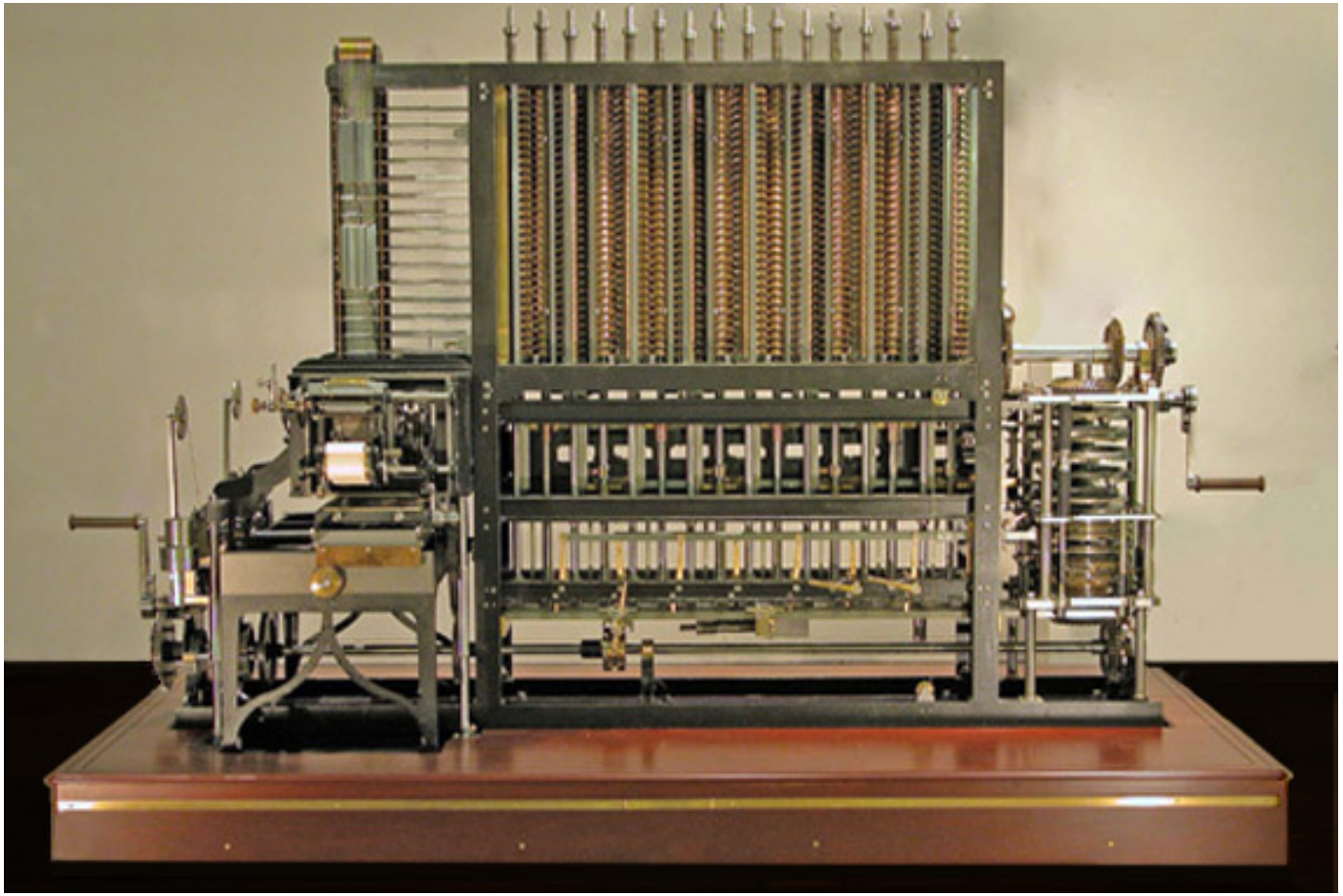


Jacquard's Loom

Also see http://www.computersciencelab.com/ComputerHistory/HistoryPt2.htm

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Difference engine



http://www.youtube.com/watch?v=0anIyVGeWOI

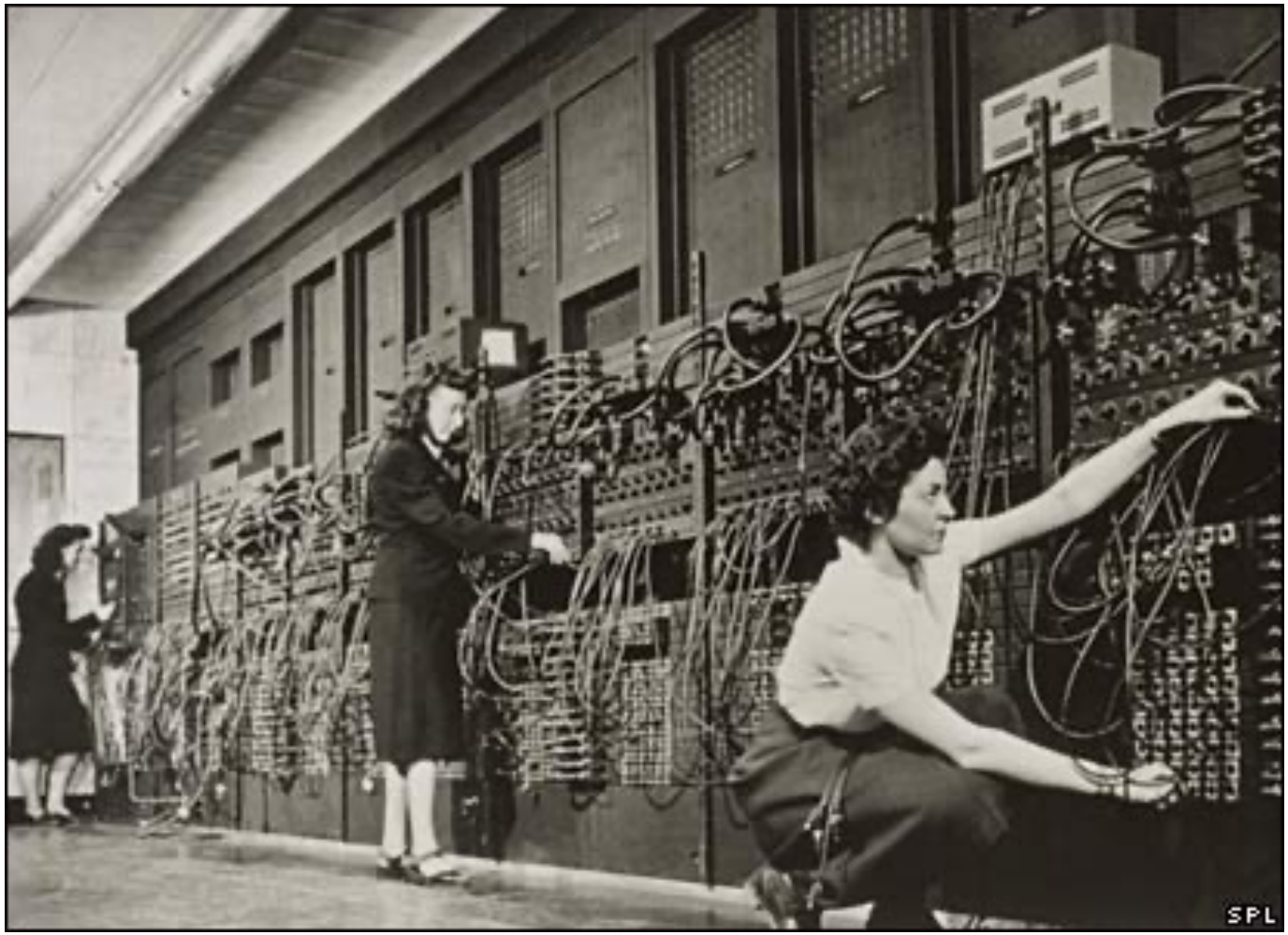
Slide from “Introduction to Computing”

The Harvard Mark-I



Grace M. Hopper working on the Harvard Mark-I, developed by IBM and Howard Aiken. The Mark-I remained in use at Harvard until 1959, even though other machines had surpassed it in performance, providing vital calculations for the navy in World War II.

Slide from “Introduction to Computing”



Programming the ENIAC

Slide from “Introduction to Computing”



History of Computation

- Assignment to be announced today:
  - Reading material
  - A video
- Quiz next week!